Programmable Logic Devices (PLDs)

Focus on PLAs and PALs

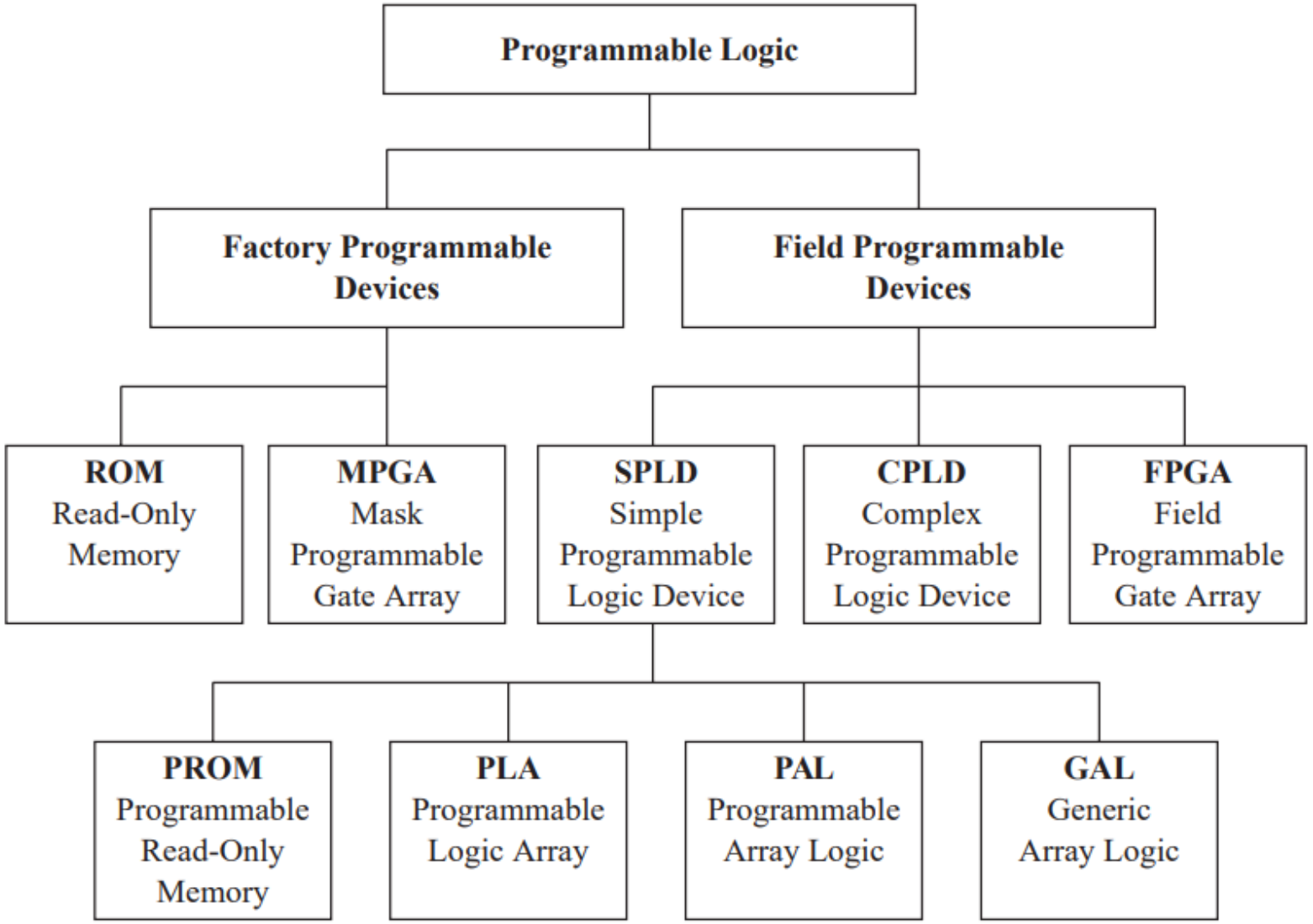
Introduction to PLDs

- Programmable Logic Devices (PLDs) are digital devices used to build reconfigurable digital circuits. They allow designers to implement logic functions without using fixed logic gates.

Fixed logic versus Programmable Logic

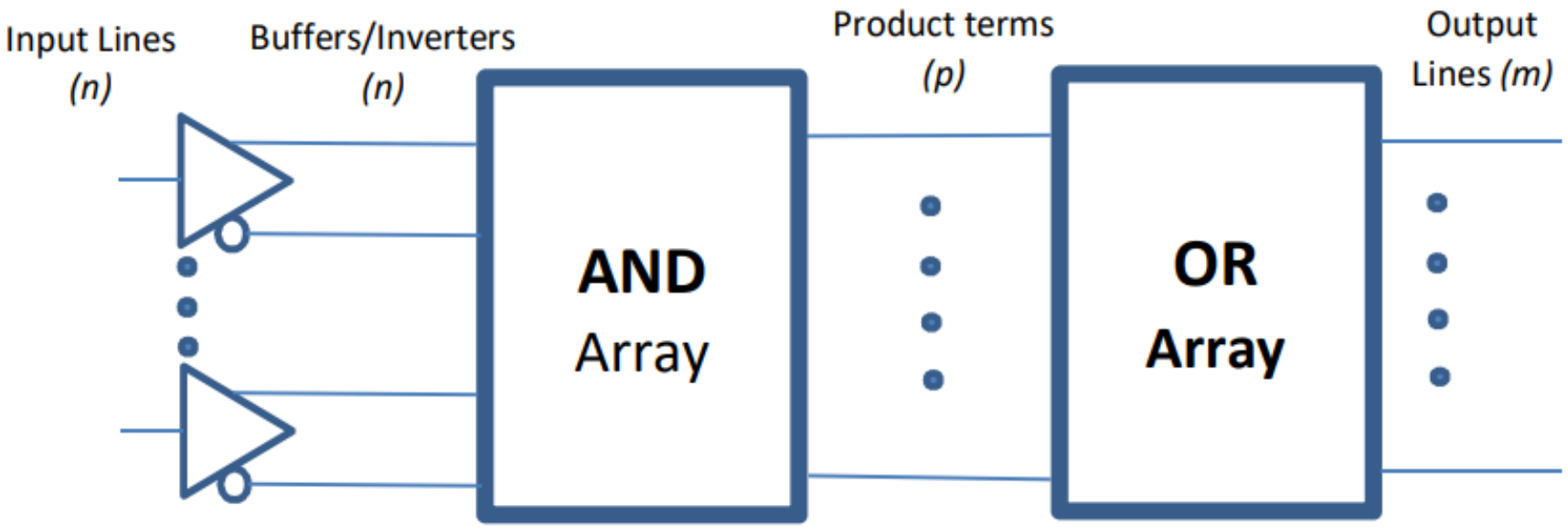
Fixed logic devices	Programmable logic devices.
These are fixed functions ICs/Devices like Multiplexers, Demultiplexers, adders, Counters, etc.	These are Application Specific integrated Circuits (ASICs) and the outcome can change accordingly.
The circuit configurations in the fixed logic system are permanent. They have designed to carry out a predetermined set of actions. The logic cannot be altered after it has been constructed and coded. For routine activities, this method is a great advantage.	Application Specific Integrated Circuits (ASICs) are produced by an IC manufacturer according to the user's specifications. ASICs are being designed and created by the designer to fulfil the unique requirements of a user. Hence, the designer aims to give an adaptable, practical, and economical solution for the user.
One tiny mistake in the manufacturing process like uploading the wrong code in the device, and the entire system is discarded, and a new design is developed. That's quite some risk that companies aren't willing to take unless necessary.	It is easy-to-program, affordable and equipped with better features. Inexpensive software is used to develop, code and test the required design.
Fixed logic does not allow the users to expand or build on their existing functionalities.	A programmable logic seems more feasible and beneficial.
Fixed function ICs have the following advantages over PLDs • Compact circuitry • Higher switching speed	ASICs have many advantages like • Lower quantity production costs • Design security • Reduced power requirement • Higher density • Reduced space requirement

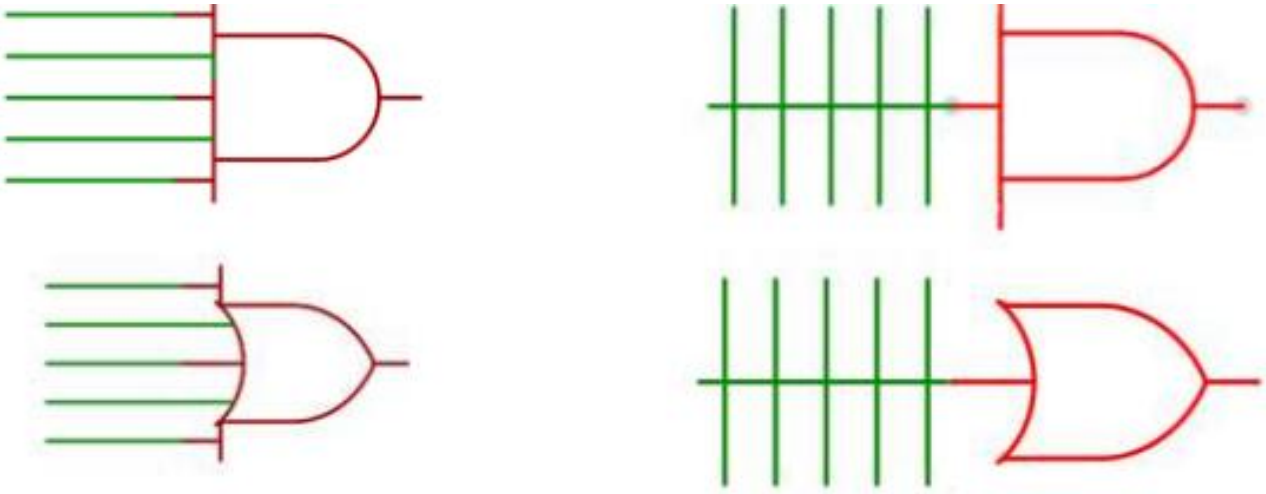
MAJOR PROGRAMMABLE LOGIC DEVICES



	SPLD	CPLD	FPGA
Density	Low Few hundred gates	Low to Medium 500 to 12,000 gates	Medium to High 3,000 to 5,000,000 gates
Timing	Predictable	Predictable	Unpredictable
Cost	Low	Low to Medium	Medium to High
Major Vendors	Lattice Semiconductor Cypress AMD	Xilinx Altera	Xilinx Altera Lattice Semiconductor Actel
Example Device Families	**Lattice Semiconductor** GAL16LV8 GAL22V10 **Cypress** PALCE16V8 **AMD** 22V10	**Xilinx** CoolRunner XC9500 **Altera** MAX	**Xilinx** Virtex Spartan **Altera** Stratix **Lattice** Mach ECP **Actel** Accelerator

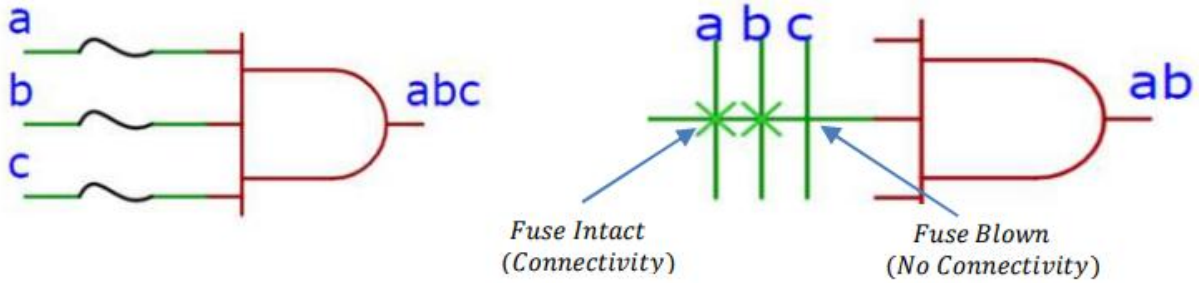
General structure of PLDs.





(a) (b)

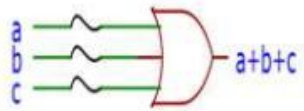
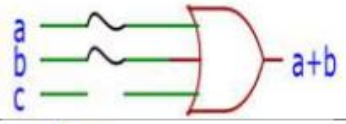
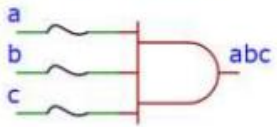
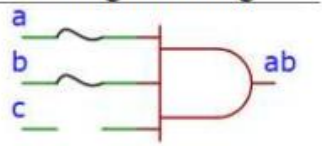
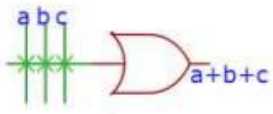
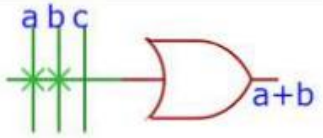
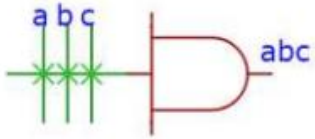
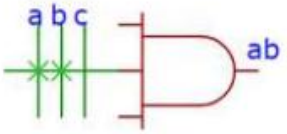
logic symbol for multiple input AND gate and OR gate (a) Conventional Symbol (b)array Logic Symbol



(a)



(b)

OR-PLD Notations		AND-PLD Notations	
Before Programming	After Programming	Before Programming	After Programming
			
			

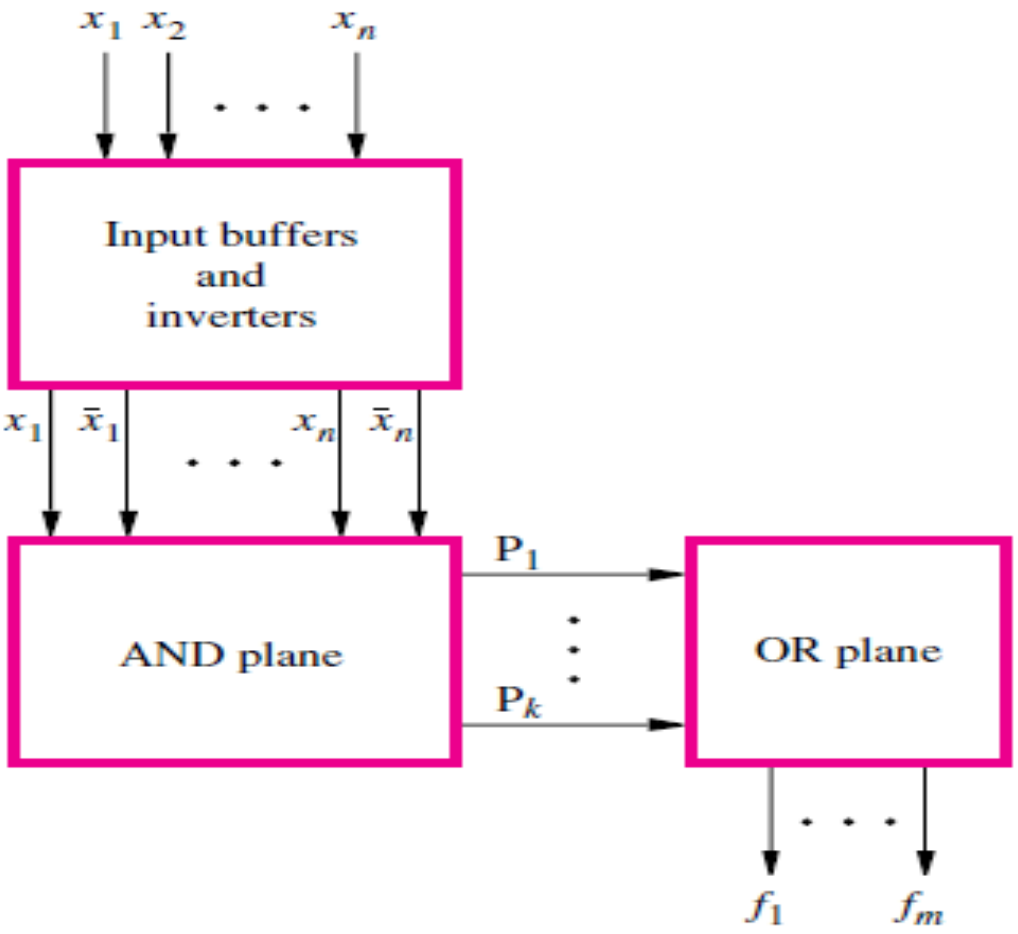
Types of PLDs

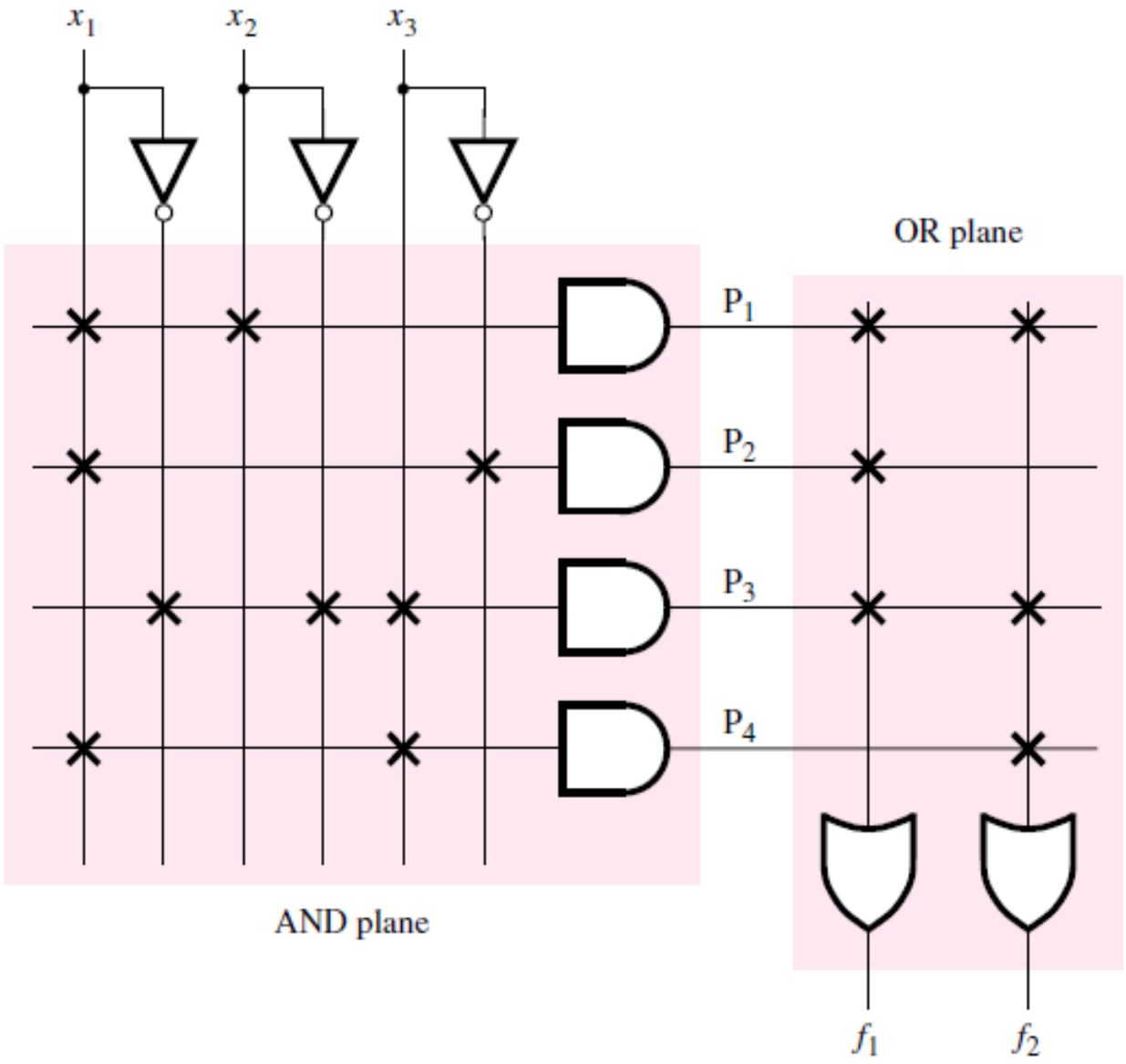
- PLDs are categorized into:
- - Programmable Logic Arrays (PLAs)
- - Programmable Array Logic (PALs)
- - Complex Programmable Logic Devices (CPLDs)
- - Field Programmable Gate Arrays (FPGAs)

Programmable Logic Arrays (PLAs)

- PLAs consist of a programmable AND array followed by a programmable OR array.
- They are used to implement complex combinational logic circuits.
- PLAs offer flexibility but are more expensive and slower compared to PALs.

Architecture of PLAs




x_1
x_2
x_3
OR plane
P_1
P_2
P_3
P_4
AND plane
f_1
f_2

$$F_0 = \Sigma m(0, 1, 4, 6) = A'B' + AC'$$
$$F_1 = \Sigma m(2, 3, 4, 6, 7) = B + AC'$$
$$F_2 = \Sigma m(0, 1, 2, 6) = A'B' + BC'$$
$$F_3 = \Sigma m(2, 3, 5, 6, 7) = AC + B$$

PLA TABLE

Product Term	Inputs			Outputs			
	A	B	C	F_0	F_1	F_2	F_3
$A'B'$	0	0	—	1	0	1	0
AC'	1	—	0	1	1	0	0
B	—	1	—	0	1	0	1
BC'	—	1	0	0	0	1	0
AC	1	—	1	0	0	0	1

$$F_1 = \Sigma m(2, 3, 5, 7, 8, 9, 10, 11, 13, 15) \qquad F_1 = bd + b'c + ab'$$
$$F_2 = \Sigma m(2, 3, 5, 6, 7, 10, 11, 14, 15) \qquad F_2 = c + a'bd$$
$$F_3 = \Sigma m(6, 7, 8, 9, 13, 14, 15) \qquad F_3 = bc + ab'c' + abd$$

If we implement these reduced equations in a PLA, a total of eight different product terms (including c) are required.

Instead of minimizing each function separately, we need to minimize the total number of rows in the PLA table.

$$F_1 = a'bd + abd + ab'c' + b'c$$
$$F_2 = a'bd + b'c + bc$$
$$F_3 = abd + ab'c' + bc$$

Reduced PLA Table

a	b	c	d	F_1	F_2	F_3
0	1	—	1	1	1	0
1	1	—	1	1	0	1
1	0	0	—	1	0	1
—	0	1	—	1	1	0
—	1	1	—	0	1	1

Example: Implement using PLA

$F1(A,B,C)=\sum m(0,1,6,7)$

$F2(A,B,C)=\sum m(2,3,4,5)$

Programmable Array Logic (PALs)

- PALs consist of a programmable AND array followed by a fixed OR array.
- They are faster and cheaper than PLAs but offer less flexibility.
- PALs are suitable for simpler logic functions.

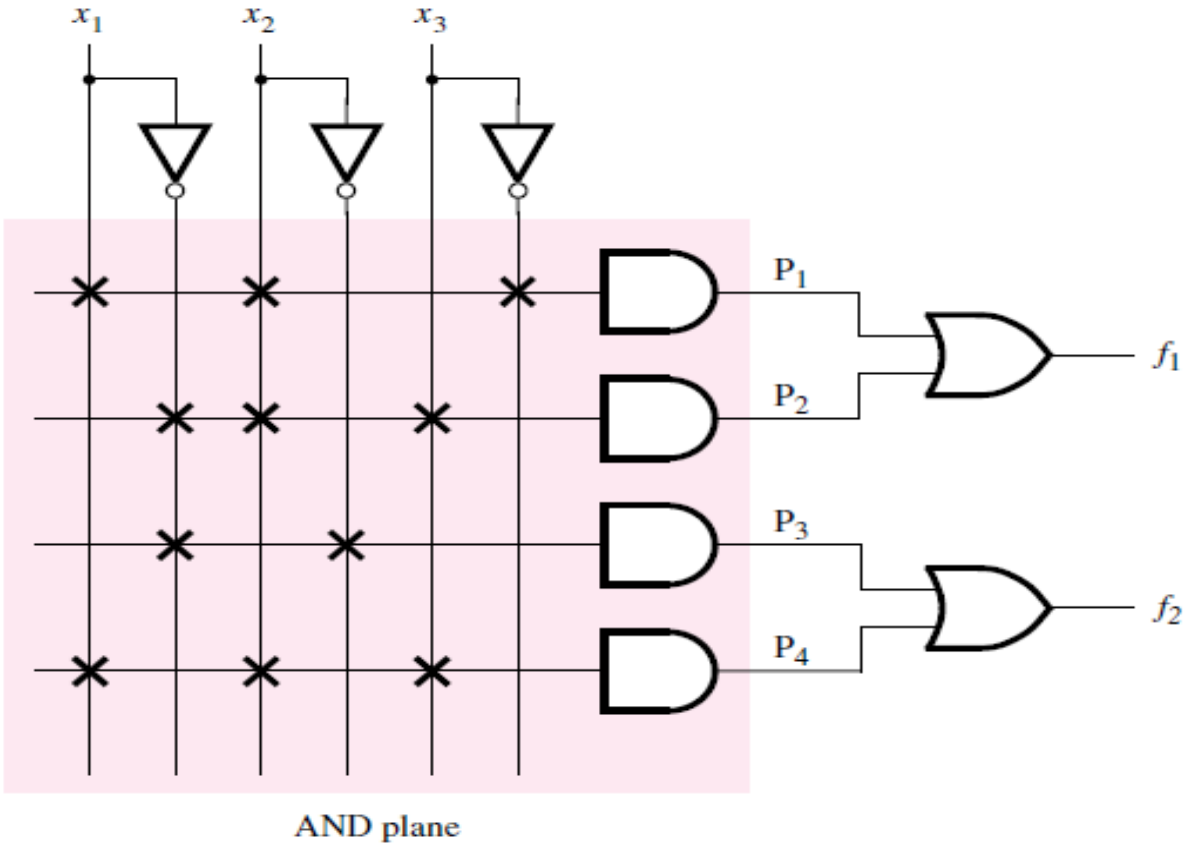
Architecture of PALs

- PAL architecture includes:
- - Input buffer
- - Programmable AND array
- - Fixed OR array
- - Output buffer

Programmable Array Logic (PAL)

The PLAs are hard to fabricate and the speed-performance of circuits implemented in the PLAs are reduced because both AND and OR planes are programmable. These drawbacks led to the development of a similar device in which the AND plane is programmable, but the OR plane is fixed. Such a chip is known as a programmable array logic (PAL). They are simpler to manufacture, and thus less expensive than PLAs, and offer better performance.

An example of a PAL with three inputs, four product terms, and two outputs :



f1 = x1x2x3’ +x1’x2x3

f2 = x1’x2 +x1x2x3.

Example: Implement using PAL

$F1(A,B,C)=\sum m(0,1,6,7)$

$F2(A,B,C)=\sum m(2,3,4,5)$

Advantages and Disadvantages

- Advantages:
- - Reconfigurable logic
- - Reduced development time
- - Cost-effective for small-scale production

- Disadvantages:
- - Limited complexity
- - Speed limitations
- - Higher cost for large-scale production

Applications of PLDs

- PLDs are used in:
- - Digital signal processing
- - Communication systems
- - Embedded systems
- - Consumer electronics
- - Industrial automation

Conclusion

- PLDs, including PLAs and PALs, play a crucial role in digital circuit design.
- They offer flexibility and efficiency for implementing logic functions.
- Understanding their architecture and applications is essential for modern electronics.